

Mohammad Mahmudul Hasan

Project Lead, AI Engineer | Geospatial AI, 3D Urban Modeling & Point Clouds

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RESEARCH FOCUS

Advancing **3D Urban Modeling** and **Autonomous GIS** by integrating Geospatial AI (GeoAI) with high-fidelity reconstruction: **Autonomous Digital Twins** (3D Gaussian Splatting & CityGML), **Spatial Intelligence** (point cloud analytics, sub-pixel camera pose estimation, and automated BIM extraction), and **Multi-Temporal Change Detection**.

EDUCATION

Islamic University of Technology (IUT), Bangladesh — B.Sc. in Electrical and Electronic Engineering, 2017 – 2021

CGPA: **3.76 / 4.00** (First Class Honours). Selected Coursework: Project & Thesis (A+ / A+), DSP (A), Neural Networks & Fuzzy Logic (A+).

Thesis: Quantifying Locomotive Features in EEG of Impaired Consciousness (Coma) with Distinctive Cerebral Rhythms. Analyzed EEG time-series to isolate discriminative spectral and temporal features characterizing depths of consciousness.

Notre Dame College, Dhaka — Higher Secondary Certificate (Science), 2014 – 2016. GPA: **5.00 / 5.00** (Government Scholarship).

PROFESSIONAL EXPERIENCE

HawarIT Limited, Bangladesh — Project Lead, AI Department (2024–Present) | Senior AI Engineer (2023–2024) | ML Engineer (2021–2023)

Lead multidisciplinary teams building AI and geospatial automation systems across photogrammetry, point clouds, and GIS platforms:

- **3D Point Cloud Intelligence & BIM Extraction:** Developed semantic segmentation and automated pipe/geometry extraction on aerial, mobile, and indoor LiDAR datasets (95% extraction accuracy, 60% manual modeling time reduction).
- **360° Street-View GeoAI Localization:** Built an end-to-end YOLOv5 and tracking pipeline detecting and localizing 130 traffic signs and 33 street furniture assets from vehicle 360° imagery (>95% detection accuracy, 92% geospatial localization precision).
- **Multi-Temporal Change Detection:** Engineered deep learning models detecting building extensions, construction, and demolitions from multi-year orthophotos; integrated into GIS web validation platform (88% accuracy, 65% manual inspection reduction).
- **3D CityGML Modeling & Texture Projection:** Built automated 3D CityGML reconstruction and occlusion-aware multi-view texture projection from aerial imagery, maintaining texture displacement < 3 pixels.
- **Orthophoto Automation & Camera Pose Estimation:** Created AI image QA (cloud/shadow/blur detection, 97% accuracy), PostGIS/AHN LiDAR seamline optimization (65% edit reduction), and camera pose estimation (2px accuracy, 90% manual AT reduction).

PEER-REVIEWED PUBLICATION

Tasnim Sakib Apon, **Mohammad Mahmudul Hasan**, Abrar Islam, and Md. Golam Rabiul Alam. “Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI.” 2021 IEEE Asia-Pacific Conference on CSDE.

doi:10.1109/CSDE53843.2021.9718400.

HONORS & SELECTED PROJECTS

Awards: Excellent Performer (2024) & Borsho Shera (Best Performer of the Year 2022), HawarIT · 2nd Place, International Innovation Tender (Sentinel Earth Observation) · 1st Place, MATLAB Competition, AUST · Top 15, Dhaka.AI Deep Learning Competition.

Earth Observation: Sentinel multispectral AI for crop classification, water quality, and algae bloom detection. **Fault Detection:** Accelerometer signal processing in MATLAB. **Astrophysics:** WMAP CMBR primordial signal isolation and sky-map generation.

TECHNICAL SKILLS & REFERENCES

Skills: Python, SQL, C, MATLAB | PyTorch, TensorFlow, OpenCV, YOLO, FastAPI | GDAL, GeoPandas, PostGIS, QGIS, CityGML | Open3D, PCL, COLMAP, Gaussian Splatting.

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